



Exploratory Data Analysis (EDA)

Techniques for Summarizing & Visualizing Datasets to Uncover Patterns, Spot Anomalies & Test Hypotheses

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WHAT IS EDA?

The "Detective" of Data Science

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Definition

EDA is the process of examining a dataset to understand its structure, uncover patterns, spot anomalies, and test assumptions — *before* building any model.

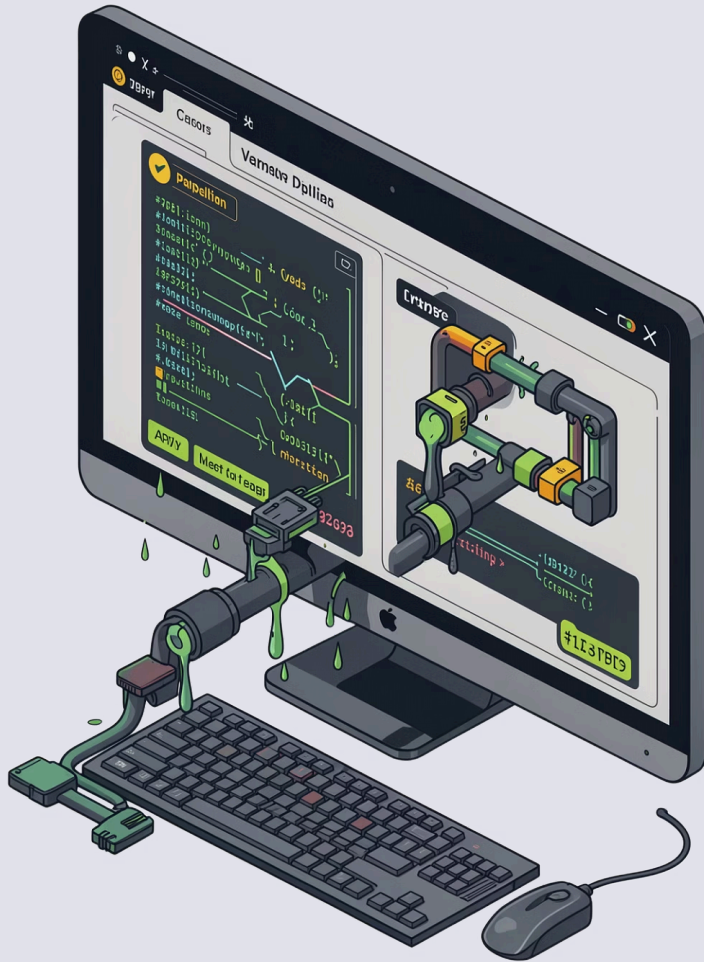
Origin

Introduced by mathematician **John Tukey in the 1970s** as a "look at the data first" approach — encouraging statisticians to explore before they conclude.

Simple Analogy

Think of EDA as a **doctor's check-up before prescribing medicine** — you examine the patient (data) thoroughly before deciding the treatment (model).

Why EDA? — What Happens Without It



✗ Without EDA

You may build a model on dirty, incomplete, or misleading data — leading to **wrong predictions** and costly decisions.

✓ With EDA

Catch **missing values, outliers, and wrong data types** early — saving both time and cost before any model is trained.

🎯 Real Example

A student marks dataset has `age = 999` for some rows. EDA catches this impossible value before it corrupts your analysis.

🔍 Goal

Generate hypotheses, understand distributions, and decide the **right analytical approach** — EDA is data-driven hypothesis generation.

Step 1 — Summary Statistics: Numbers That Tell the Story



Mean, Median, Mode

Measure the "center" of data — e.g., average marks of 60 students = **72.5**



Standard Deviation & Variance

Measure how spread out the data is — high SD means marks vary widely across students.



Min, Max, Range

Spot extreme values instantly — e.g., marks range from **12 to 98**.



Python: `df.describe()`

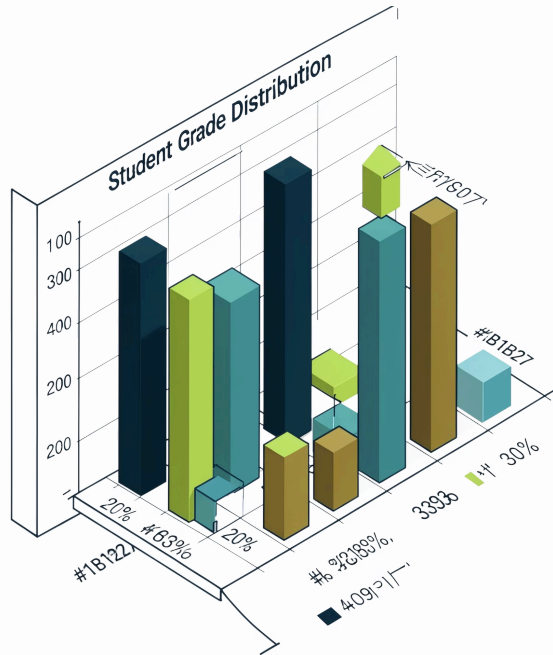
Gives count, mean, std, min, 25%, 50%, 75%, max in **one single command**.

Summary Statistics — Student Marks Dataset

Statistic	Marks (out of 100)
Mean	72.5
Median	74
Std Dev	12.3
Min	12
Max	98

📄 `df.describe()` — run this one line in Python to instantly get all statistics above for every column in your dataset.

Step 2 — Univariate Analysis: One Variable at a Time



1

Histogram

Shows **frequency distribution** — e.g., how many students scored 60–70, 70–80, etc. Reveals whether data is bell-shaped, skewed, or bimodal.

2

Box Plot

Displays **median, quartiles, and outliers** in one chart — a student scoring 12 appears as a dot outside the whisker.

3

Outlier Rule (IQR Method)

Any value below **Q1 – 1.5×IQR** or above **Q3 + 1.5×IQR** is automatically flagged as an outlier.

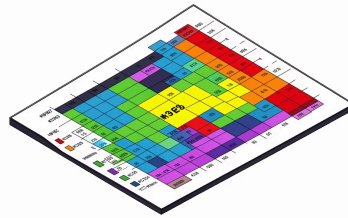
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Step 3 — Bivariate & Multivariate Analysis: Finding Relationships



Scatter Plot (Bivariate)

Plot two variables — e.g., **Study Hours vs. Marks**. A rising trend shows positive correlation between the two variables.



Correlation Matrix & Heatmap

A table of correlation values (-1 to $+1$) between all variable pairs. **Dark red = strong positive; dark blue = strong negative** correlation.



Real Example

In a house price dataset, **area and price show correlation = 0.87** — a strong positive relationship confirmed visually by the heatmap.

Step 4 — Spotting Anomalies & Testing Hypotheses

Z-Score Method

Values beyond ± 3 **standard deviations** from the mean are automatically flagged as anomalies.

IQR Outlier Method

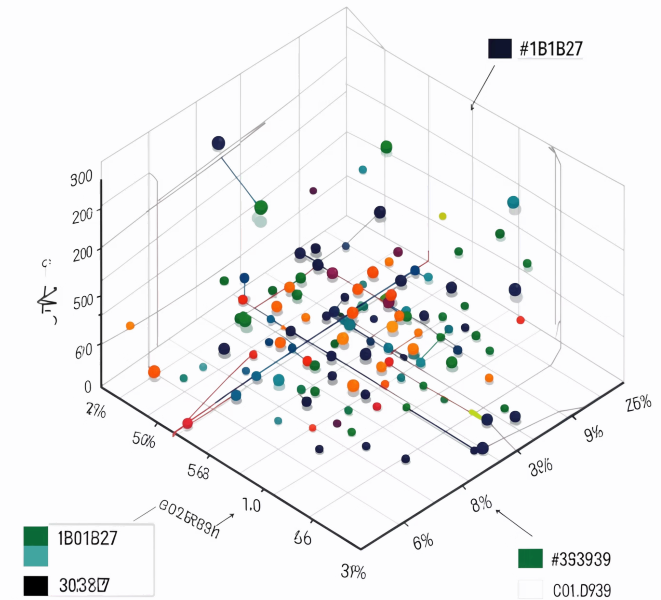
Use $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ thresholds to detect and remove extreme values reliably.

Missing Value Heatmap

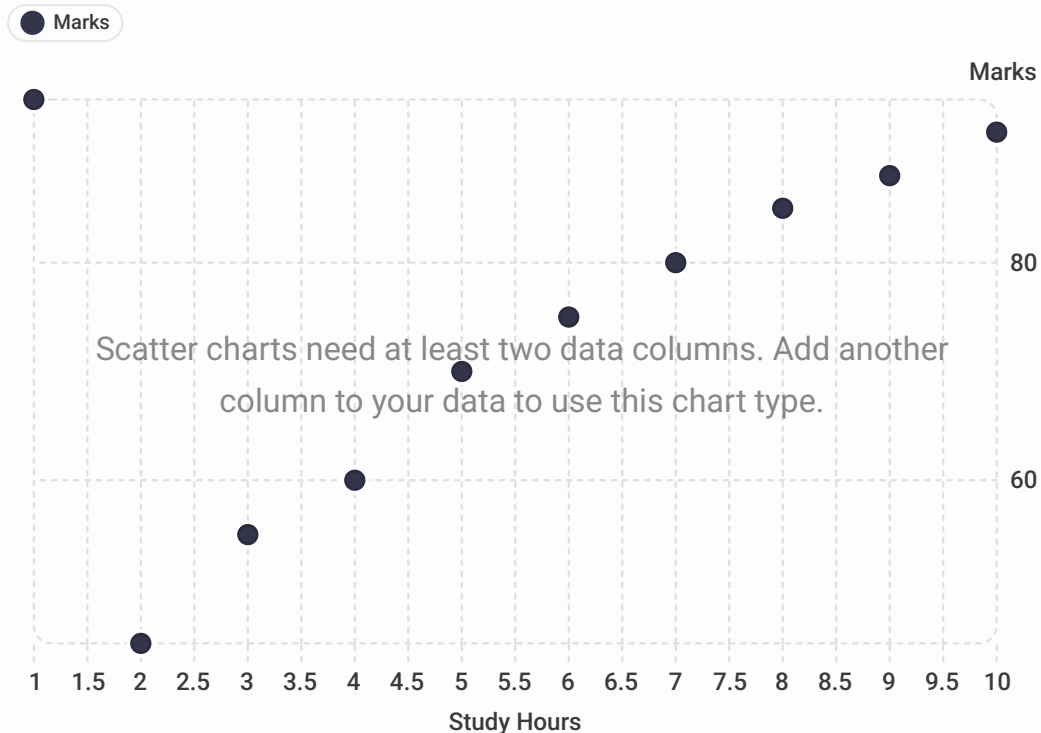
Run `sns.heatmap(df.isnull())` to visually map all null/missing cells across your entire dataset.

EDA Generates Hypotheses

EDA *does not confirm* hypotheses — it **generates** them. Example: "Do students who study >5 hrs score above 80?"



Histogram & Scatter Plot — Detecting Patterns, Trends & Anomalies



Reading This Scatter Plot

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→ Positive Trend

As Study Hours (X) increase, Marks (Y) rise — **upward slope = positive correlation.**

→ Anomaly Spotted!

The point at **(1 hr, 95 marks)** lies far from the cluster — a data error or exceptional student worth investigating.

→ Ice Cream Analogy

Scatter plot of temperature vs. ice cream sales shows strong positive trend; a **winter spike is an anomaly** worth investigating.

Correlation Analysis — Identifying Relationships Between Variables

What is Correlation?

A number between **-1 and +1** that measures how strongly two variables move together.

Pearson's r formula:

$$r = \frac{\sum[(X-\bar{X})(Y-\bar{Y})]}{\sqrt{[\sum(X-\bar{X})^2 \times \sum(Y-\bar{Y})^2]}}$$

Python: `df.corr()` + `sns.heatmap()`

⚠ Correlation ≠ Causation — Ice cream sales and drowning both rise in summer, but one doesn't cause the other!

Correlation Matrix — House Price Dataset

Variable	Area	Age	Rooms	Price
Area	1.00	-0.12	0.75	0.87
Age	-0.12	1.00	-0.30	-0.65
Rooms	0.75	-0.30	1.00	0.80
Price	0.87	-0.65	0.80	1.00

Area vs Price = +0.87 (strong positive); Age vs Price = -0.65 (moderate negative — older house, lower price)

+1

Perfect Positive

0

No Relation

-1

Perfect Negative

Hypothesis Testing — Validating Assumptions with Statistics

What is a Hypothesis?

A testable assumption about data — e.g., *"Students who study >5 hrs score above 70 on average."*

Result Example

$p = 0.003$ $\rightarrow p < 0.05 \rightarrow$
Reject $H_0 \rightarrow$ Studying >5
hrs *does* significantly
improve marks.

Simple Rule

$p < 0.05$ = "Yes, the
difference is real, not by
chance." Always interpret
in context of your
problem.

01

State H_0 (Null)

Average marks ≤ 70 for all students
regardless of study hours.

02

State H_1 (Alternate)

Students studying >5 hrs score > 70 on
average.

03

Choose $\alpha = 0.05$

5% chance of error — the standard
significance threshold in statistics.

04

Run t-test

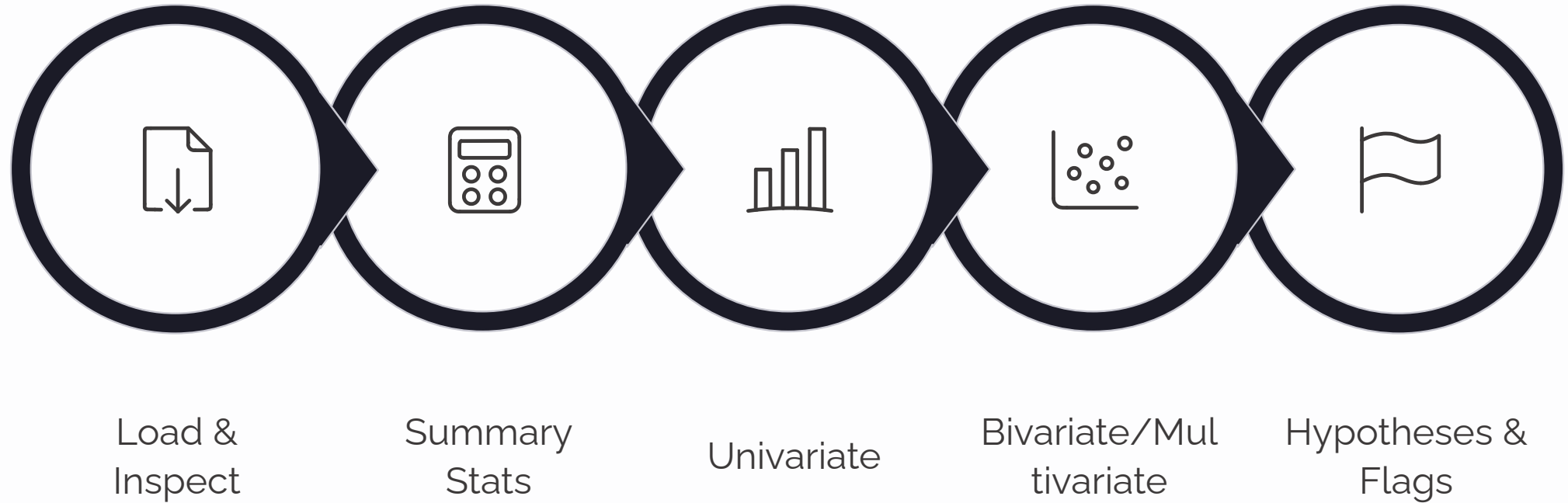
```
scipy.stats.ttest_ind(group1, group2)
```

05

Interpret p-value

If **$p < 0.05$** , reject H_0 — the result is statistically significant!

EDA Workflow — Putting It All Together



This end-to-end EDA pipeline ensures that every dataset is thoroughly understood before any machine learning model is applied. Each step builds on the previous, progressively revealing the data's structure, distributions, relationships, and anomalies.

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Key Takeaways — EDA in a Nutshell

Always EDA First

Never skip EDA — it is the foundation of every reliable data science project.
Garbage in = garbage out.

Summary Stats → Distributions → Relationships

Follow the natural progression:
describe → visualize univariate →
explore bivariate/multivariate.

Visuals Are Powerful

Histograms, box plots, scatter plots,
and heatmaps reveal what numbers
alone cannot — always plot your data.

EDA Generates Hypotheses

EDA doesn't prove anything — it *generates* testable hypotheses for formal statistical validation.

Correlation ≠ Causation

Always question relationships critically — ice cream and drowning both peak in summer, but neither causes the other!

Sources & References

LatentView Analytics

What Is Exploratory Data Analysis?
Types, Techniques, and Steps

Wikipedia

Exploratory Data Analysis –
Wikipedia

Penn State STAT 508

2 Exploratory Data Analysis (EDA) –
Applied Data Mining and Statistical
Learning

Coursera

What Is an Exploratory Data Analysis
(EDA)? Types, Uses, and Tools

Modern Statistics with R

4 Exploratory Data Analysis and
Unsupervised Learning | Modern
Statistics with R